Pavel Mosharev

Physicist and algorithm researcher | Quantum and quantum-inspired optimization

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Profile

Current work studies quantum and quantum-inspired algorithms for discrete optimization through mechanism-focused analysis and reproducible numerical experiments. Background spans theoretical physics, university teaching, and industrial algorithm R&D at Huawei.

Professional Experience

2025–Present — **Postdoctoral Researcher** — **USTC**
University of Science and Technology of China, School of Physical Sciences (Hefei, Anhui, China)
- Study quantum and quantum-inspired discrete optimization through Ising/QUBO models, variational quantum algorithms, energy landscapes, and controlled numerical experiments.
- Develop collaborative research with quantum-science groups in Shenzhen; contribute to experiment design, validation, manuscripts, and student research advising.

Sep 2025–Mar 2026 — **Visiting Scholar** — **GBA Quantum Science Center**
Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area, Shenzhen, China; six-month concurrent appointment.

2023–2025 — **Postdoctoral Research Fellow** — **Huawei**
Huawei Technologies Co., Ltd., 2012 Laboratories, Shenzhen, China
- Researched quantum and quantum-inspired optimization through solver simulation and benchmarking, heuristic analysis, Ising-model construction and preprocessing, and related SAT/MILP investigations.
- Developed protected-routing algorithms for network resilience as the main applied line; also contributed to wireless optimization. Two algorithms were transferred to a Huawei product line; received the Future Star Award 2024.

2022–2023 — **Algorithm Engineer** — **Huawei**
Huawei Russian Research Institute, Moscow, Russia
- Developed quantum-inspired constrained-optimization methods and protected-routing algorithms for optical/IP networks; received the Huawei Spark Award 2022.

2020 — **Independent Contractor / Research Project** — **Gazprom Neft**
Gazprom Neft PJSC, Moscow, Russia
- Delivered Python tools to assess CCE/PVT laboratory data and estimate reservoir-fluid saturation pressure.

2016–2022 — **Assistant, later Senior Lecturer** — **MPEI**
National Research University "Moscow Power Engineering Institute" (MPEI), Moscow, Russia
- Taught university mathematics, including analysis, linear algebra, differential equations, and probability.

Feb–Jun 2021 — **Senior Lecturer** — **Gubkin University**
Gubkin University, Moscow, Russia; taught general physics and supervised laboratory practice.

Sep–Dec 2021 — **Instructor** — **HSE University**
HSE University, Moscow, Russia; led undergraduate Linear Algebra seminars for economics students.

Education

2016–2021 — **PhD in Physics (Candidate of Sciences)** — **MSU**
Lomonosov Moscow State University, Moscow, Russia

2010–2016 — **Specialist in Physics, with honours** — **MSU**
Lomonosov Moscow State University, Moscow, Russia

Technical Skills

- **Algorithms and optimization:** Discrete and combinatorial optimization; graph algorithms; constrained and multi-objective optimization; solver benchmarking.
- **Quantum and quantum-inspired methods:** Variational quantum algorithms/QAOA; Ising/QUBO/HUBO modeling and reduction; simulated bifurcation and quantum-annealing-inspired algorithms; circuit simulation.
- **Scientific computing:** Python for numerical modeling, data analysis, visualization, and validation; Qiskit and MindQuantum; C++ (selected projects).
- **Research engineering:** Reproducible experiments, testing and provenance; Git; Linux/WSL and remote computing workflows; AI-assisted coding and research workflows with coding agents.

Languages

- Russian — Native
- English — Advanced (IELTS Academic 7.5, 2024)
- Chinese — Upper-intermediate (HSK 5, 2026)

Selected Research and Technical Contributions

- **Quantum optimization and algorithm behavior:** Work combines theoretical analysis with exact simulations and energy-landscape studies to understand how shallow-circuit and related algorithms explore discrete problems.
- **Industrial graph optimization:** Developed cost-driven pruning methods for constrained protected routing and led their later reproducible evaluation on public network benchmarks.
- **Higher-order Ising modeling:** Co-conceived a reduction method for general-order Ising models and co-developed its code and numerical experiments; a minimal public reference implementation accompanies the work.
- **Applied scientific computing:** Built Python methods for CCE/PVT data processing, quality checks, and saturation-pressure estimation in a Gazprom Neft research project.

Selected Publications

1. X. Zhu, Z. Zou, F. Jin, **P. Mosharev**, et al. "Combinatorial optimization enhanced by shallow quantum circuits with 104 superconducting qubits." *National Science Review*, 13(9), nwag124, 2026. Equal contribution.
2. **P. A. Mosharev**, C.-M. Lee, X. Shu, X. Zhang, and M.-H. Yung. "Cost-Driven Prunings for Iterative Solving of Constrained Routing Problem With SRLG-Disjoint Protection." *IEEE Transactions on Networking*, 34, 92–106, 2026.
3. C. Mao, **P. Mosharev**, Y. Wang, and M.-H. Yung. "A reduction scheme for general-order Ising-like Hamiltonians in quantum heuristic solvers." *Quantum Information Processing*, 25(10), 319, 2026.
4. G. Kalachev, **P. Mosharev**, Z. Zou, P. Panteleev, and M.-H. Yung. "Pilot-Wave Simulator: Exact Classical Sampling from Ideal and Noisy Quantum Circuits up to Hundreds of Qubits." *Quantum*, 10, 2173, 2026. Corresponding author.
5. X.-Z. Tao, **P. Mosharev**, and M.-H. Yung. "Multi-Objective Optimization by Quantum-Annealing-Inspired Algorithms." arXiv:2604.26477, 2026. Preprint; manuscript submitted; equal contribution.
6. Q.-G. Zeng, X.-P. Cui, B. Liu, Y. Wang, **P. Mosharev**, and M.-H. Yung. "Performance of quantum annealing inspired algorithms for combinatorial optimization problems." *Communications Physics*, 7, 249, 2024.
7. A. A. Lobanov, **P. A. Mosharev**, et al. "Mathematical analysis of the behavior of PV isotherms of reservoir fluids." *Geology, Geophysics and Development of Oil and Gas Fields*, 8(380), 39–50, 2023 (in Russian; title translated).
8. O. V. Kechkin and **P. A. Mosharev**. "A General Harmonic Solution in Dilaton Electrodynamics: An Exact Expression for the Fields and the Generalized Lorentz Force." *Moscow University Physics Bulletin*, 75(5), 427–433, 2020.